

So far | Row opⁿ: $R_i + a R_j$, $R_i \leftrightarrow R_j$, $c R_i$ ($c \neq 0$) all reversible

Using row op^s, any matrix $\xrightarrow[\text{pass}]{\text{forward}}$ REF $\xrightarrow[\text{pass}]{\text{backward}}$ RREF

REF: pivots move to the right as you go down (all 0 rows at bottom)

RREF: entries above pivots are also 0. Location of pivots stays same!

Use for solving $Ax=b$ $\rightsquigarrow$ Augmented matrix $[A|b]$ corresponding to
col^s of A w/o pivots

Consistent $\iff$ REF($A|b$) has no pivot in last column

If consistent, then {solutions} $\longleftrightarrow$ arbitrary value assignment to "free variables"

REF($A|b$) helps us write any pivot variable x_i uniquely in terms of x_j 's with $j > i$

RREF($A|b$) (... // ...) free vars x_j with $j > i$

$$Ax=0 \sim [A|0]$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \end{bmatrix} = \begin{bmatrix} x_1 \\ -ax_3 - px_5 - sx_6 \\ x_3 \\ -qx_5 - tx_6 \\ x_5 \\ x_6 \\ 0 \\ 0 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ -a \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 0 \\ -p \\ 0 \\ -q \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_6 \begin{bmatrix} 0 \\ -s \\ 0 \\ -t \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 & x_8 \\ 0 & 1 & a & 0 & p & s & 0 & 0 \\ 0 & 0 & 0 & 1 & q & t & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Free vars. x_1, x_3, x_5, x_6

Defⁿ in $\mathbb{R}^n$ = $\left\{ \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \mid a_i \in \mathbb{R} \right\}$ we call elts of $\mathbb{R}^n$ (column) vectors $\vec{0} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$
(denoted $u, v, w, \dots$)

① Given vectors $v_1, \dots, v_k$ in $\mathbb{R}^n$ and scalars $a_1, \dots, a_k$ the vector $a_1 v_1 + \dots + a_k v_k$ is called a linear combination of $v_1, \dots, v_k$ (with wts $a_1, \dots, a_k$)

Given a subset S of $\mathbb{R}^n$, span of $S :=$ set of all lin. comb. of vectors in S
ie. $\text{Span}(S) = \{ a_1 v_1 + \dots + a_k v_k \mid a_i \in \mathbb{R}, v_i \in S \}$. (convention: empty sum = $\vec{0}$ vector)

② A subspace of $\mathbb{R}^n$ is a nonempty set $W \subset \mathbb{R}^n$ that is closed under vector addition and multiplication by any scalar. i.e. $\left. \begin{array}{l} w_1, w_2 \in W \\ a \in \mathbb{R} \end{array} \right\} \Rightarrow \begin{array}{l} w_1 + w_2 \in W \\ a w_1 \in W \end{array}$

How do subspaces arise? In two natural ways

Examples : " 0 " = $\{0\}$ and $\mathbb{R}^n$ are subspaces of any $\mathbb{R}^n$

nonzero proper subspaces of $\mathbb{R}^1$: none

——//—— of $\mathbb{R}^2$: lines thro' origin i.e. $\mathbb{R}v = \{av \mid a \in \mathbb{R}\}$ for a fixed $v \neq 0$

——//—— of $\mathbb{R}^3$: lines $\{av \mid a \in \mathbb{R}\}$ and "planes" $\{av + bw \mid a, b \in \mathbb{R}\}$ with fixed nonproportional v & w

① For any subset S of $\mathbb{R}^n$ $\text{span}(S)$ is a subspace of $\mathbb{R}^n$

② The set of solutions of a homogeneous system $Ax = 0$ i.e. $\{w \mid Aw = 0\}$
is a subspace of $\mathbb{R}^c$. (where A is $r \times c$)

Given $r \times c$ matrix A , consider the function $f(x) = Ax$ $f: \mathbb{R}^c \rightarrow \mathbb{R}^r$

Observe ① $f(v_1 + v_2) = f(v_1) + f(v_2)$

② For any scalar c , we have $f(cv) = cf(v)$

③ Ex. Any function from $\mathbb{R}^c \rightarrow \mathbb{R}^r$ satisfying ① & ② must be of the form $f(x) = Ax$ for a magic $r \times c$ matrix A .

When is $f(x) = Ax$ injective? surjective?

Get equivalent conditions in terms of

Important
Exercise

- solutions of a linear system

- cols of A

- $\text{RREF}(A)$

- rows of A

Defⁿ Such functions are called linear transformations